

Polarization-resolved Second Harmonic Imaging of collagen organization in connective tissues – Application to biomechanics

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Abstract: Polarization-resolved SHG microscopy enables measurements of the main orientation and the angular dispersion of collagen fibrils within the focal volume. Results are presented in cornea, tendon and skin, eventually as function of mechanical stress.

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1. Introduction

Collagen is the main component of connective tissues, including skin, cornea, bone, arteries... and is responsible for their biophysical and mechanical properties. This structural protein is synthesized as a triple helix, which self-assembles into fibrils (10-300 nm diameter) and further forms various three-dimensional (3D) organizations specific to each tissue [1]. *In situ* 3D visualization of this biopolymer is therefore a major biomedical concern in order to probe its accumulation or disorganization in many diseases, or to guide the engineering of tissue substitutes with a suitable 3D organization.

Second harmonic generation (SHG) microscopy has proved to be a powerful technique for the visualization of fibrillar collagen [2, 3]. It offers high specificity and unequalled contrast, without any prior labeling, which is not possible using conventional imaging techniques. Further information about the 3D organization of collagen can be gained with polarization-resolved second harmonic generation (P-SHG) microscopy. Two quantitative structural parameters are measured by rotating the linear polarization of the exciting laser beam: mean in-plane orientation of collagen molecules in the excitation volume ($0.350 \times 0.350 \times 1.0 \mu\text{m}^3$) and SHG signal anisotropy that is the ratio of SHG signal for parallel versus perpendicular orientation of the excitation polarization to the collagen main direction.

The aim of this paper is to demonstrate which biological information about collagen 3D distribution can be obtained from P-SHG microscopy. Illustration will be given in collagen-rich tissues which exhibit different degrees of order: rat-tail tendon (crimped fibers composed of aligned fibrils with about 100-200 nm diameter), human cornea (aligned 30nm diameter fibrils within 2-3 μm -thick stacked lamellae forming a plywood structure) and murine skin (disordered fibrils with various diameters). Application to biomechanics will be discussed.

2. Experimental setup

P-SHG microscopy is performed by mean of a rotating half waveplate inserted at the back pupil of the objective lens. Ellipticity of the excitation polarization is corrected by mean of a quarter waveplate. No analysis of the SHG polarization is performed in most of our experiments in order to obtain robust results even in scattering tissues. All the experimental parameters are kept constant during a set of P-SHG experiments to enable reliable comparison between different samples or for the same sample during a mechanical assay. It has indeed been shown that the collection direction and the numerical aperture (NA) of the excitation and collection can affect the measurement of the SHG signal anisotropy [4, 5]. Measurement of this parameter may also be affected in anisotropic tissues because of the birefringence and the diattenuation experienced by the excitation beam [6]. This issue is addressed by post-processing the depth profiles of P-SHG data in thick tissues or by recording P-SHG data only in thin tissues.

3. Results and discussion

Analytical calculations using the tensorial formalism of nonlinear optics in the plane-wave approximation show that two quantitative parameters can be extracted from the P-SHG data: the main orientation of the collagen fibrils within the focal plane, and the SHG signal anisotropy. The latter parameter is obtained from fitting or Fourier Transform of P-SHG data and amounts to the ratio of SHG signal for parallel versus perpendicular orientation of the excitation

polarization to the collagen main direction (see Figure 1). It reflects the distribution of the peptide bonds within the focal volume considering that these peptide bonds are the elementary “harmonophores” responsible for the SHG signal of fibrillar collagen [7]. In collagen-rich tissues exhibiting well-aligned fibrils in the focal plane, the SHG signal anisotropy therefore measures the angle θ of the nonlinear dipoles to the triple helix axis and is close to the direction of the peptide bonds along the triple helix [7, 8]. Measurements in human cornea and in stretched rat-tail tendon provide a value of 1.36 that corresponds to $\theta=50^\circ$ [9, 10]. Nevertheless, in disordered tissues, measurement of the SHG anisotropy parameter shows higher values that reflects the disorder of the fibril distribution within the focal volume (see Figure 1). This is demonstrated by combining P-SHG microscopy with traction assays and increasingly stretching a rat-tail tendon to align the fibrils [9]. Alternatively, higher SHG anisotropy parameter may correspond to well-aligned fibril out of the focal plane. This is demonstrated in collagen liquid crystals mimicking the corneal lamellar organization [11]. All these considerations are confirmed by numerical simulations in a highly focused regime [6, 11].

P-SHG microscopy is then combined to biomechanical assays to monitor the reorganization of the collagen distribution upon controlled mechanical stretching. In rat-tail tendon, as mentioned above, we observe a decrease of the SHG signal anisotropy revealing an increased alignment of the collagen fibrils at sub-micrometer scale [9]. In Human cornea, inflation assays show no variation of the SHG anisotropy parameter, indicating that there is no reorganization of collagen fibrils within the lamellae. In contrast, variation of the distribution of the lamellae directions at the scale of the whole cornea is observed, which indicates an inter-lamellae reorganization upon pressure. Importantly, the direction of the lamellae is only obtained from P-SHG experiments, while the basic SHG images show a homogeneous contrast, with no hint for lamellar direction, due to the small diameter of the cornea fibrils: 30 nm, below the optical resolution. Finally, a reorganization of the collagen fibrils direction towards the stretching direction is also observed in murine skin upon traction [12].

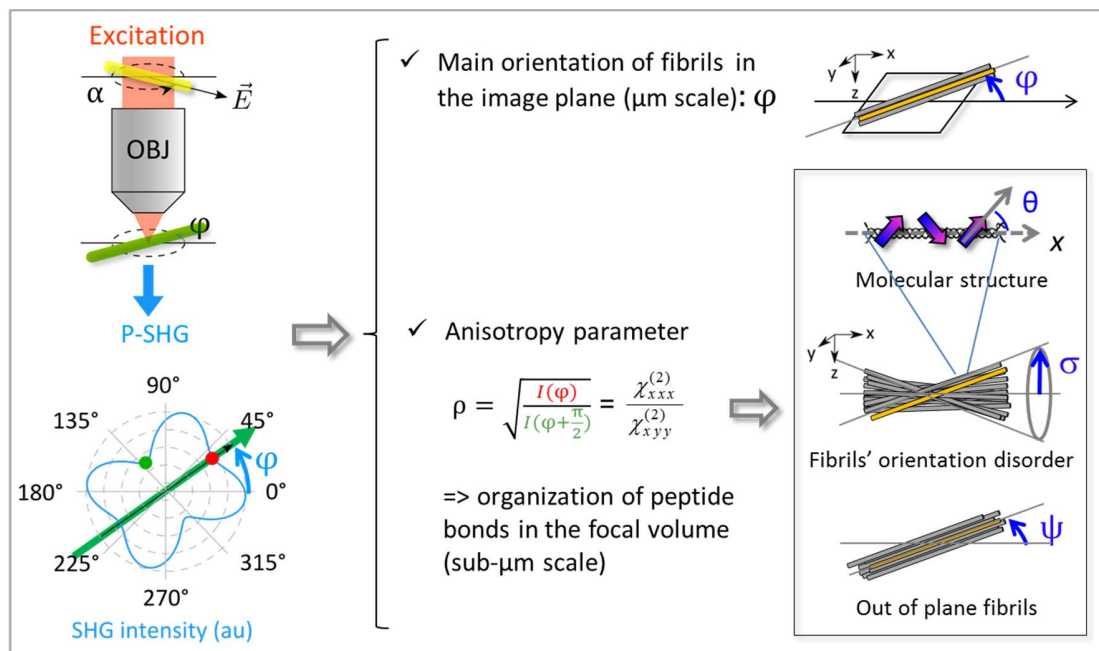


Figure 1. P-SHG experimental setup and processed parameters. Rotation of the linear polarization of the laser excitation enables to record polarimetric diagrams and to extract 2 quantitative parameters: the main orientation of the collagen fibrils within the focal plane (micrometer scale), and the anisotropy parameter that is sensitive to the organization of the peptide bonds within the focal volume (sub-micrometer scale).

4. Conclusion

All these experimental results prove the efficiency of P-SHG microscopy for *in situ* characterization of the 3D distribution of collagen in tissues and biomimetic matrices. We have shown that the SHG signal anisotropy is related to the molecular structure of collagen and to its 3D distribution inside the excitation volume at sub-micrometer scale: it increases with the orientation disorder and with the out-of-plane orientation angle [1, 2]. P-SHG can then monitor the increase of the alignment of collagen fibrils in a tendon or in a skin biopsy subjected to controlled mechanical stretching and decipher the 3D structure of collagen liquid crystals, in particular the out-of-plane orientation of collagen

molecules [1]. We have also illustrated the sensitivity of P-SHG microscopy to anisotropic structures at sub-micrometer scale by imaging corneas that are composed of stacked lamellae containing aligned fibrils with 30 nm diameter: P-SHG images reveals the direction of these collagen fibrils while they are not resolved in the SHG images [4]. Finally, P-SHG can be combined to biomechanical assays to monitor the reorganization of fibrillar collagen microstructure upon macroscopic mechanical stress.

5. References

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